NEXUS 360° AEC COMMAND OS

The Multi-Tenant AI-Powered Engineering ERP, Field Supervision & FIDIC Claims Platform Built for Multidisciplinary Mega-Consultancies

MISR CONSULT (MC #35) + DEC READY

ISO 9001 • 14001 • 45001

Unifying 527+ Engineers, 6 Core Disciplines & 316 Field Supervisors in One Real-Time Command Center

Purpose-built for House of Expertise #35 (**Prof. Dr. Sameh Abdel-Gawad**) & Dar El-Khaleej: connecting Cairo HQ design studios with remote WWTP, SWRO desalination, and infrastructure sites across Egypt, KSA, and the UAE.

TOTAL WORKFORCE	MEGA-PORTFOLIO	AI ARCHITECTURE
527 Staff	**$1.4B+**	**100% Sovereign**
316 Site • 128 Design • 83 Support	WWTP • SWRO • Bridges • Silos	Local Ollama LLM + Cloud Sync

3 PILLARS OF EXECUTIVE VALUE

1. Multi-Discipline RIBA & BOQ Control
Live tracking across Water/SWRO, Civil/Structural, MEP/SCADA, Architecture/BIM, Bridges, and Geotech.

2. GPS-Geofenced Field Companion OS
Mobile NCR/Snag photo pins, Daily Site Diaries, and IR approvals synced directly from remote sites.

3. FIDIC Claims & 360° EPI Scoring
Automated Sub-Clause 8.4 / 20.1 EOT dossiers and objective 4-factor Engineer Performance Index.

LIVE PRODUCTION DEPLOYMENT
https://milestonetracker.web.app

LIVE NOW

Fragmented Silos Across HQ Studios & Remote Sites

Why WhatsApp photos, disconnected Excel BOQs, and manual FIDIC claim drafting cost mega-consultancies millions in unrecovered variations

01

The Field-to-HQ Blindspot

316+ site engineers supervise remote WWTPs, pump stations, and highways. Site photos, NCRs, and daily diaries sit trapped in WhatsApp threads and Excel sheets instead of feeding live RIBA & BOQ progress.

Result: 5–10 Day Reporting Lag

02

6-Discipline Coordination Chaos

Process/Hydraulic, Civil/Structural, Electromechanical/SCADA, BIM/Arch, Bridges, and Geotech teams work in disconnected trackers—causing interdisciplinary clashes and late submittals.

Result: Submittal Bottlenecks

03

Slow, Manual FIDIC EOT Claims

When utility clashes, client variations, or sovereign delays hit, assembling a defensible **FIDIC Red/Yellow/Silver Book** notice takes weeks of forensic document hunting—risking the 28-day time bar.

Result: Unrecovered EOT & Fees

04

Subjective Engineer Evaluation

Evaluating 500+ staff relies on subjective manager memory rather than objective CAD workstation telemetry, GPS site attendance, submittal pass rates, and milestone velocity.

Result: Hidden Underperformance

68% of EOT & variation claims are weakened by incomplete contemporaneous site records

14+ Hrs wasted weekly per project director reconciling Excel S-curves with site WhatsApp reports

100% Solvable when CAD telemetry, GPS site logs, BOQs, and FIDIC AI live in one unified database

How NEXUS Fixes Every Bottleneck Immediately

Replacing fragmented spreadsheets and chat threads with a unified, multi-tenant AEC Command Center

BEFORE NEXUS (LEGACY WORKFLOW)

- ❌ **Disconnected Excel Trackers:** Weekly manual S-curve updates that are already outdated by Monday morning.
- ❌ **Unverified Site Attendance:** Paper sign-in sheets and unindexed WhatsApp site photos across 20+ sites.
- ❌ **Manual FIDIC Claim Drafting:** 3 weeks of legal/contract compilation per delay event; high risk of missed 28-day notices.
- ❌ **Public Cloud AI Leaks:** Engineers pasting confidential tender specs into consumer ChatGPT accounts.

WITH NEXUS 360° AEC OS (LIVE TODAY)

- ✅ **Live RIBA 0–7 & Earned Value BI:** Real-time CPI/SPI, BOQ burn vs. physical progress, and 6-discipline radar charts.
- ✅ **GPS-Geofenced Field Companion:** 1-tap NCR photo pins, weather/manpower logs, and instant HQ escalation.
- ✅ **1-Click FIDIC EOT Claim Dossiers:** Automated Sub-Clause 8.4, 13.1 & 20.1 legal narratives with quantum breakdown.
- ✅ **Air-Gapped Local AI (Ollama):** Run DeepSeek-R1 & Qwen2.5-Coder on-premise with zero external data leakage.

10x Faster
FIDIC EOT & Variation Dossier Generation

100% Verified
Geofenced Site Supervision & CAD Telemetry

Zero Leaks
On-Premise Ollama LLM + Tenant Isolation

< 60 Sec
White-Label Firm Provisioning & Demo Seeding

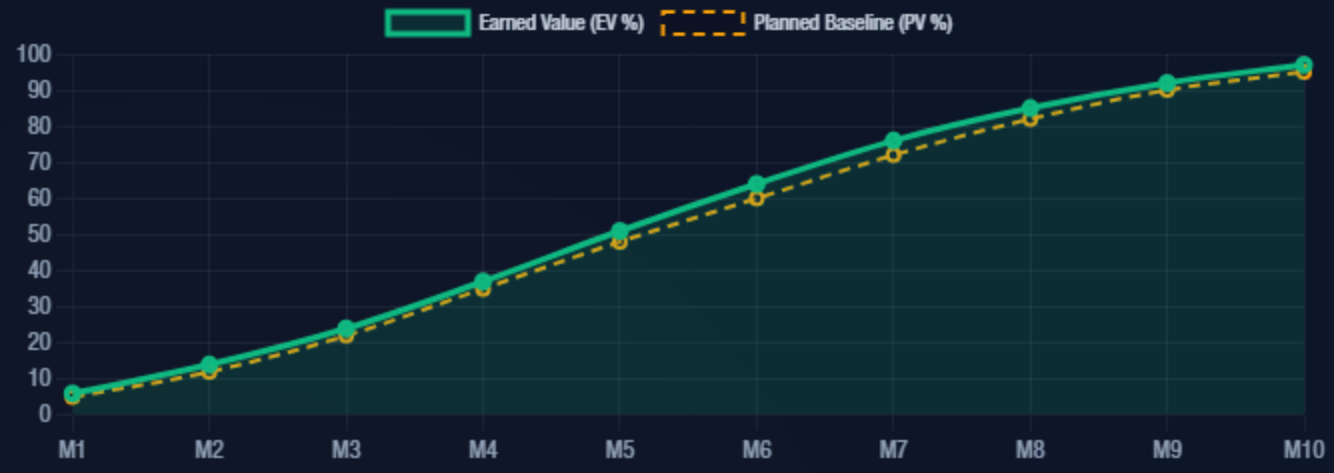
One Dual-Mode Platform: Portal + NEXUS ERP

Switch seamlessly between Executive Consultancy Portal and full Turnkey Engineering ERP across Desktop, Tablet, and Mobile

LIVE PORTFOLIO S-CURVE (EARNED VALUE VS. PLANNED VS. ACTUAL)

Aggregated across WWTP, SWRO Desalination, Bridges & Grain Silos

SPI 1.04 • CPI 1.02



8 INTEGRATED MODULES IN ONE APP

1. Executive BI & S-Curve
2. 6-Discipline Hub
3. 360° EPI Scoreboard
4. Live CAD & Site Telemetry
5. Field Companion Mobile
6. FIDIC EOT Claim Engine
7. Commercial Risk & HSE
8. Boardroom TV Mode

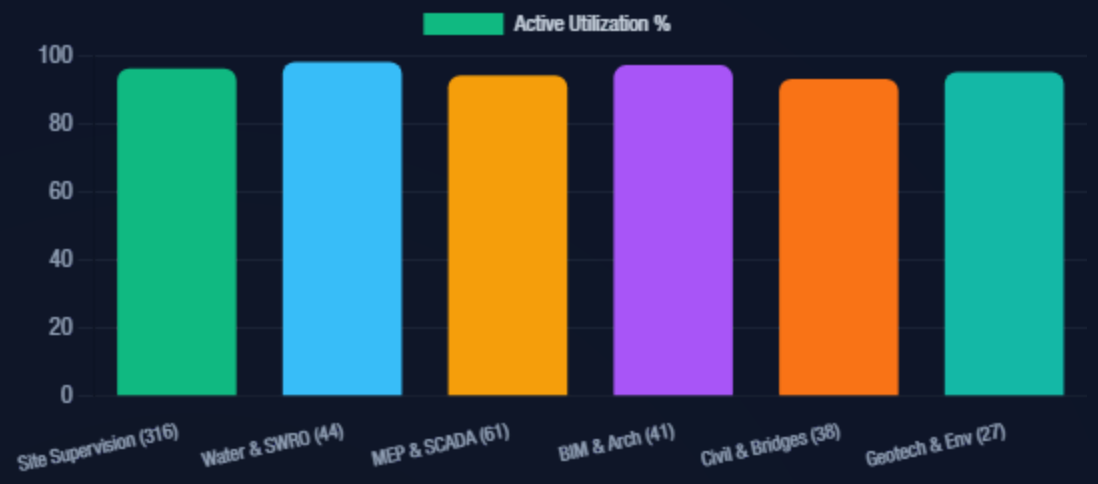
MOBILE + BOARDROOM RESPONSIVE

Works identically on a **65-inch Cairo Boardroom screen** (with auto-cycling TV mode) and on a **375px iPhone screen** in the middle of a Toshka or Sinai construction site.

6-Discipline Engineering, RIBA 0–7 & BOQ Control

Purpose-built for Misr Consult's exact organizational structure (527 Staff across Water/SWRO, Civil, MEP, BIM, Bridges & Supervision)

MISR CONSULT 527-STAFF HEADCOUNT & UTILIZATION BY DIVISION



LIVE MEGA-PROJECT PORTFOLIO PRE-LOADED FOR MC (#35)

Project / Client	Scope & Capacity	RIBA Stage	Health
Abu Rawash Mega WWTP	1.6M m³/day Sludge & Civil	Stage 5 • Construction	96%
Marafek / DEC SWRO (UAE)	60,000 m³/d Desalination	Stage 4 • Tech Design	98%
Toshka & Sinai Supervision	316 Field Staff • Pumping/Canals	Stage 5 • Site Works	91%
22 Strategic Grain Silos	730,000 Tons Slip-Form Civil	Stage 6 • Handover	99%

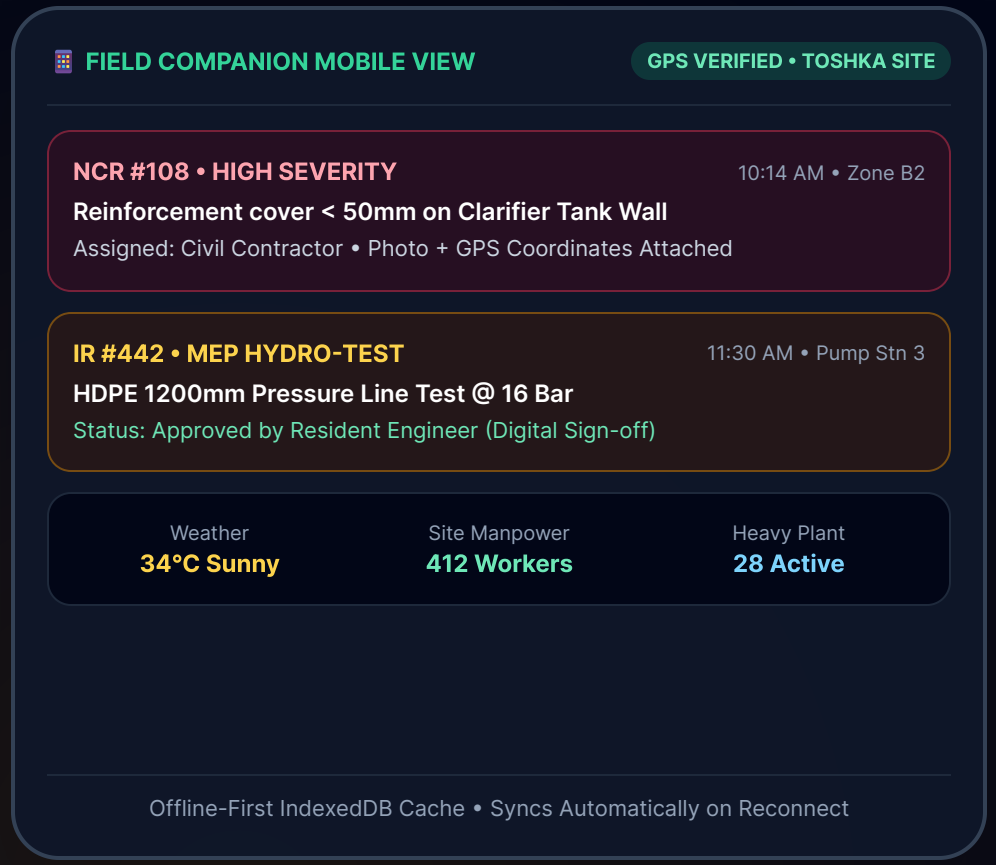
SUBMITTAL FIRST-TIME PASS
94.8%

ACTIVE REVIT / CIVIL3D SYNC
BIM LOD 400

ISO STAGE-GATE AUDIT
100% Compliant

Mobile Site Supervision, Geofenced GPS & NCR Pinning

Empowering 316+ Resident Engineers & Site Inspectors across remote sites with instant mobile-to-HQ synchronization



How Field Companion Transforms Site Supervision

1. Geofenced GPS Check-In

Verifies resident engineers are physically inside the project boundary (Toshka, New Delta, Abu Rawash, UAE) with live lat/long telemetry.

2. Instant NCR & Snag Photo Pinning

Capture defect photos, tag discipline (Civil, MEP, SWRO, Safety), assign severity, and lock contractorpour approvals until resolved.

3. Contemporaneous FIDIC Evidence

Every daily site diary, weather delay, and obstructed access log is timestamped and feeds directly into the FIDIC EOT Claim Generator.

4. Deep Workstation + Field Sync

Combines Cairo studio CAD/BIM active hours with remote site GPS logs in the Live Operations Monitor.

IMPACT FOR MISR CONSULT'S 316 SUPERVISION STAFF

Eliminates 100% of manual daily report typing and creates an unshakeable audit trail for ISO 9001 & 45001.

0 Lag

Automated FIDIC EOT Dossiers & 360° Engineer Scoring

Protecting consultancy fee margins under FIDIC Red/Yellow/Silver Books while ranking top engineering talent objectively

⚖️ **FIDIC EOT & VARIATION CLAIM GENERATOR** — 1999 / 2017 Editions

1-Click Legal Dossier & Quantum Calculation

Select any delay event (e.g., *Unforeseen Subsurface Utilities at Toshka Canal* or *SWRO High-Pressure Pump Customs Delay*) and automatically generate a ready-to-sign claim notice:

- **Contractual Basis:** Auto-cites FIDIC Sub-Clause 20.1 (28-Day Notice), 8.4 (EOT), 4.12 (Unforeseeable Physical Conditions) & 13.1 (Variations).
- **Critical Path & Quantum:** Calculates net critical delay days, site supervision prolongation cost, and remobilization fees.
- **Contemporaneous Exhibits:** Links directly to timestamped Field Companion diaries, NCRs, and RFIs.

Plus: ISO 45001 HSE TRIR & Monte Carlo Risk Matrix — **Export PDF / Print Ready**

🏆 **360° ENGINEER PERFORMANCE INDEX (EPI)** — 4-Pillar Weighted Formula

35%	25%	25%	15%
CAD/BIM Telemetry	Geofenced Site GPS	Submittal QA Pass	Milestone Velocity

#1 Eng. Ahmed Seera (Mega WWTP Division) — **100 / 100**
CAD: 100% • GPS: 100% • QA: 98% • Logs: 100%

#2 Eng. Ahmed Abdel-Gawad (SWRO & Marafek UAE) — **99 / 100**
CAD: 100% • GPS: 99% • QA: 97% • Logs: 99%

#3 Eng. Mohamed Saber (Site Supervision – 316 Staff) — **97 / 100**
CAD: 100% • GPS: 96% • QA: 93% • Logs: 95%

Responsive 2-tier cards ensure zero text truncation on both desktop and mobile viewports.

100% Sovereign Local AI (Ollama) + Real-Time Firestore

Zero data leakage for sovereign water, military, and infrastructure tenders—powered by on-premise LLMs

STAGE 1 • MULTI-SOURCE INGESTION

Unified Telemetry & Project Context

Workstation Daemons: Active Revit, AutoCAD, Civil 3D, WaterGEMS, and Primavera P6 window telemetry.

Field Companion GPS: Mobile check-ins, NCR defect pins, and daily manpower/weather logs.

Multi-Tenant Firestore: Strict tenant-isolated real-time collections (`firms / projects / logs`).

Real-Time Snapshot Sync (<150ms)

STAGE 2 • SOVEREIGN AI PROCESSING

Local Ollama LLM + RAG Pipeline

100% On-Premise Ollama: Connects to `localhost:11434` running **Qwen2.5-Coder 32B**, **DeepSeek-R1 32B**, or **Llama 3**.

Live Context Injection: Injects active project KPIs, overdue milestones, SPI/CPI variances, and FIDIC clauses into every prompt.

Zero-Latency Fallback: Built-in deterministic AEC domain engine if offline on remote sites.

Air-Gapped Sovereign Security

STAGE 3 • EXECUTIVE ACTION OUTPUTS

Instant Decisions & Legal Dossiers

AI Engineering Copilot: Answers *"Summarize delayed projects"* or *"Draft FIDIC EOT notice"* in seconds.

Automated EPI Calculation: Recomputes 360° engineer rankings dynamically as logs arrive.

Boardroom TV Broadcast: Auto-cycles 5 executive views every 12 seconds for client visits.

Exportable to PDF, CSV & Print

Launch Your Firm Workspace in Under 60 Seconds

Zero 6-month ERP implementation pain—how a new consultancy goes from login to live FIDIC & BI output in 3 steps

STEP 1 • 15 SECONDS Provisioning

Launch Tenant or One-Click Demo

Visit `milestonetracker.web.app/login`. Click **"Launch Misr Consult (#35) Demo"** or **"+ New Firm"** to provision an isolated multi-tenant workspace pre-seeded with your divisions, projects, and staff.

Included Out of the Box:
- Custom Firm Name & Theme
- Role-Based Access (Admin/Lead/Engineer)
- Pre-Seeded Mega-Project Portfolio

STEP 2 • 20 SECONDS Live Operations

Track RIBA, EPI & Field Logs

Land on the **Executive Command Center**. Filter the **360° EPI Scoreboard** by discipline (Water & SWRO, Site Supervision, MEP, BIM, Civil), or log a mobile site inspection in **Field Companion**.

Instant Visibility:
- Earned Value S-Curve (CPI / SPI)
- 6-Discipline Submittal QA Pass Rates
- Live CAD & Geofenced GPS Telemetry

STEP 3 • 25 SECONDS AI & Claims

Generate FIDIC Claim or Board Mode

Open the **FIDIC Claim Engine** to export a formal Sub-Clause 8.4/20.1 EOT dossier, ask the **AI Copilot** for a portfolio risk summary, or launch **Boardroom TV Mode** for client presentations.

Immediate ROI Outputs:
- Signed FIDIC EOT Notice Dossier
- ISO 45001 HSE & Risk Register
- Full-Screen Auto-Cycling TV Mode

Zero Disruption Adoption: Can run alongside existing Primavera P6 / Revit workflows on Day 1 via CSV/Excel import & workstation telemetry.

Time to Value: < 1 Hour

3 High-Impact Scenarios Built for Misr Consult & DEC

How Principals, Resident Engineers, and Contract Directors use NEXUS every single day

SCENARIO 1 • EXECUTIVE BOARDROOM

Chairman & Managing Principals (Prof. Dr. Sameh Abdel-Gawad)

During a ministerial review or Marafek UAE partner visit, leadership switches on **Boardroom Presentation TV Mode**. The 65" screen auto-cycles live KPIs, S-Curves, 6-discipline radars, and top EPI engineers every 12 seconds.

Measurable Outcome:
Unmatched institutional prestige; proves real-time ISO 9001/14001/45001 governance to sovereign clients.

SCENARIO 2 • REMOTE MEGA-SITES

Head of Construction Supervision (316 Field Staff)

Across Toshka, New Delta, and Hayah Karima WWTP sites, resident engineers check in via **Field Companion GPS**, pin concrete cover NCRs with photos, and log daily manpower. Cairo HQ sees every active site in real time.

Measurable Outcome:
Saves 15+ hours/week per site lead and eliminates unverified attendance or lost WhatsApp defect reports.

SCENARIO 3 • COMMERCIAL PROTECTION

Contracts, Claims & Cost Control Directors

When electro-mechanical equipment customs delays or underground utility clashes impact critical path, the director opens **FIDIC Claim Generator** and exports a Sub-Clause 8.4 & 20.1 EOT dossier in 2 minutes.

Measurable Outcome:
Zero missed 28-day FIDIC time bars and full recovery of prolonged site supervision fees.

Multi-Tenant Cloud, Dedicated White-Label, or Sovereign On-Prem

Tailored commercial tiers designed for growing engineering firms and 500+ staff Houses of Expertise

TIER 1 • DIVISION PILOT

Department Cloud

Up to 50 Engineers • Instant Setup

- Multi-Tenant Isolated Firestore Workspace
- Executive BI S-Curves & RIBA 0–7 Tracker
- 360° Engineer Performance Index (EPI)
- Field Companion Mobile Web App
- Standard FIDIC EOT Claim Templates

Ideal for single-project or single-division pilot

TIER 2 • HOUSE OF EXPERTISE

RECOMMENDED FOR MC & DEC

Enterprise White-Label

500+ Staff Across Egypt, KSA & UAE

- **Full Custom Branding** (MC #35 & DEC Portals)
- Unlimited Mega-Projects & 6-Discipline Hubs
- 316+ Field Supervision Mobile Companion Seats
- Full FIDIC Red/Yellow/Silver Claim & Quantum Engine
- Boardroom TV Mode + Dedicated Onboarding

Ready for immediate 30-Day Pilot Rollout

TIER 3 • NATIONAL SECURITY GRADE

Sovereign On-Prem + AI

Air-Gapped Local Ollama LLM Cluster

- Everything in Enterprise White-Label
- **On-Premise Ollama AI** (Qwen2.5 / DeepSeek-R1)
- Zero External Cloud Data Transmission
- Custom Revit / Primavera P6 / ERP Connectors
- Dedicated SLA & Custom Feature Engineering

Built for sovereign water, silo & defense tenders

Let's Step Inside Your Live Misr Consult Command Center

Pre-seeded with MC's 527-staff structure, Abu Rawash WWTP, Marafek UAE SWRO, Toshka Supervision, and 22 Strategic Grain Silos

LIVE INTERACTIVE DEMO READY NOW

Experience Misr Consult (#35) & DEC Live in 1 Click

Open **https://milestonetracker.web.app/login** on any laptop, tablet, or phone in this room and click **"Launch Misr Consult (#35) Demo"**.

Live URL (Desktop & Mobile)
https://milestonetracker.web.app

Pre-Configured Tenant
Misr Consult (MC #35) & DEC

Prepared by: **Eng. Michael Magdy**

Engineered for: **Prof. Dr. Sameh Abdel-Gawad & MC / DEC Leadership**

PROPOSED 30-DAY PILOT ROADMAP

Week 1 • Brand & Portfolio Alignment
Configure MC (#35) & DEC custom domains, import active WWTP/SWRO BOQs, and onboard division heads.

Week 2 • Field Companion Pilot
Equip resident engineers on 2 flagship sites with mobile GPS check-in, NCR photo pinning, and daily logs.

Week 3–4 • FIDIC Claims & Board Review
Deploy local Ollama AI Copilot on-premise and review first 30-day 360° EPI & Earned Value report.

Let's Switch to the Live Platform Walkthrough →